STT 861: Theory of Probability & Statistics I

Lecture 2: Set Theory II & Counting I

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Sep 03, 2026

Reference: Casella & Berger Ch. 1.1–1.2

Lecture 2 Outline

Today

- Recap of set operations
- Symmetric difference; three-set Venn diagrams
- Counting: Fundamental theorem, ordered sampling
- Permutations

Reading: CB §1.1–1.2

Additional Set Operations

Set Difference & Symmetric Difference

Difference: $B - A = BA^c$
(elements in B but not A)

Symmetric Difference:
$B \triangle A = (B - A) \cup (A - B)$
$= (A \cup B) - (A \cap B)$
(elements in exactly one of A, B)

Numeric Example

$A = \{1, 2, 3\}$, $B = \{2, 3, 4, 5\}$
$B - A = \{4, 5\}$
$A \triangle B = \{1, 4, 5\}$

Three-set Venn diagram regions:

For events A, B, C there are $2^3 = 8$ regions:

Region	Expression
Only A	ABC^c... wait, AB^cC^c
Only B	A^cBC^c
Only C	A^cB^cC
$A \cap B$ only	ABC^c
$A \cap C$ only	AB^cC
$B \cap C$ only	A^cBC
All three	ABC
None	$A^cB^cC^c$

Recap: Partition and Containment

Key Relationships

1. $A \subset B$ iff $A = AB$ iff $B^c \subset A^c$
2. $A = B$ iff $A \subset B$ and $B \subset A$
3. $A = A \cup \emptyset = A \cap S; \quad \emptyset^c = S; \quad S^c = \emptyset$
4. $(A^c)^c = A$

Proof of (1)

Claim: $A \subset B \Leftrightarrow B^c \subset A^c$.
($\Rightarrow$): Let $x \in B^c$, so $x \notin B$. Since $A \subset B$, if $x \in A$ then $x \in B$ — contradiction. So $x \notin A$, hence $x \in A^c$.
($\Leftarrow$): Suppose $B^c \subset A^c$. Let $x \in A$. Then $x \notin A^c$, so $x \notin B^c$ (since $B^c \subset A^c$), hence $x \in B$. $\square$

Why Counting?

Classical Probability

When all outcomes are *equally likely*:

$$(A) = \frac{\#\text{ outcomes in } A}{\#\text{ outcomes in } S} = \frac{|A|}{|S|}$$

Computing $|A|$ and $|S|$ requires **counting**.

Motivating Problems

- How many ways to arrange 6 friends in a row for a photo?
- How many 4-digit PINs have no repeated digits?
- In a group of 10, how many committees of 3 are possible?
- What is P(full house in 5-card poker)?

Fundamental Theorem of Counting

Theorem 1.2.14 (CB)

If a job consists of k separate tasks, and the i-th task can be done in n_i ways $(i = 1, \ldots, k)$, then the entire job can be done in

$$n_1 \times n_2 \times \cdots \times n_k \text{ ways.}$$

Applications

- **Experiment:** 2 temperatures $\times$ 3 pressures $\times$ 4 humidities $=$ 24 combinations.
- **Passwords:** 8 characters, each from 62 possibilities (a-z,A-Z,0-9) $= 62^8 \approx 2.18 \times 10^{14}$.
- **Flip 3 coins:** $2^3 = 8$ outcomes.
- **Wardrobe:** 5 shirts $\times$ 3 pants $\times$ 2 shoes $=$ 30 outfits.

Ordered Sampling I: With Replacement

Consider a set $\mathcal{A} = \{A_1, \ldots, A_N\}$ of N distinct objects.

Ordered Sample with Replacement

An **ordered sample of size** r **with replacement** from N objects:

Each draw is from the full set (same object can be chosen twice).

$$\text{Number of ordered samples} = N^r$$

Examples

- PINs (4 digits, 0–9): $10^4 = 10{,}000$
- Braille: 6 dots, each raised or flat: $2^6 = 64$ patterns
- Toss coin n times: 2^n outcomes

Intuition

Each of the r draws independently comes from N choices. By the fundamental theorem:

$$N \times N \times \cdots \times N = N^r$$

Ordered Sampling II: Without Replacement – Permutations

Ordered Sample without Replacement

An **ordered sample of size r without replacement** from N distinct objects:

Once an object is drawn, it is not returned.

$$\text{Number of samples} = N(N-1)(N-2)\cdots(N-r+1) = \frac{N!}{(N-r)!} =: P(N,r)$$

Factorial

$N! = N \times (N-1) \times \cdots \times 2 \times 1 \quad \text{and} \quad 0! = 1.$
$P(N,r)$ is called the number of **permutations** of r from N.

Examples

- Arrange 6 friends in a row: $P(6,6) = 6! = 720$
- Seat 6 friends at a *round* table: $(6-1)! = 5! = 120$ (fix one seat)
- Choose President, VP, Secretary from 10: $P(10,3) = 10 \times 9 \times 8 = 720$

Computing Probabilities with Permutations

Bookshelf Problem (CB)

Mr. Jones has 10 books: 4 math (M), 3 chemistry (C), 2 history (H), 1 language (L). He wants all books of the same subject together. How many arrangements?

Step 1: Arrange the 4 subjects: $4! = 24$ orderings of groups.
Step 2: Within each group: $4! \times 3! \times 2! \times 1! = 24 \times 6 \times 2 \times 1 = 288$.
Total: $4! \times 4! \times 3! \times 2! \times 1! = 24 \times 288 = \mathbf{6{,}912}$.

Probability of a Specific Arrangement

If 10 books are placed randomly on a shelf, what is $P(\text{all math books together})$?

Favorable: Treat 4 math books as 1 block $\to$ 7 objects, 7! arrangements $\times 4!$ internal arrangements.
Total: 10!

$P = \frac{7! \times 4!}{10!} = \frac{5040 \times 24}{3628800} = \frac{120960}{3628800} \approx 0.0333$

Implication Notation and Sub-Events

Sub-event / Implication

$A \subset B$ means "A implies B": knowing A occurred tells us B also occurred.

Example: $A = \{\text{roll a 6}\}$, $B = \{\text{roll an even}\}$.
$A \subset B$: rolling a 6 implies rolling an even. ✓

Ordering arrangements with indistinguishable objects

Consider a objects of type A and b objects of type B (within each type, objects are indistinguishable). Number of distinct arrangements of all $a + b$ objects:

$$\binom{a+b}{a} = \frac{(a+b)!}{a!\, b!}$$

Example: Arrange 4 M's and 3 I's in a row: $\binom{7}{4} = 35$ ways.

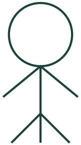
The 100 Prisoners Dilemma — The Problem

Setup

1. **100 prisoners** are each assigned a unique number 1–100.
2. **100 boxes** are placed in a room, numbered 1–100. Each box contains one slip with a *random* prisoner number (one slip per box, all shuffled).
3. Each prisoner enters the room *alone* and may open **at most 50 boxes**.
4. Prisoners **cannot communicate** after the process begins. They may agree on a strategy beforehand.
5. Goal: every prisoner finds their own number. If even one fails, all are executed.

Can they survive? Is there any hope?

The 100 Prisoners Dilemma — Illustration



Prisoner i

opens ≤ 50 boxes

goal: find slip i

$\longrightarrow$

Box 1	**Box 2**	**Box 3**
3	**6**	**1**
Box 4	**Box 5**	**Box 6**
5	**2**	**4**

6 boxes shown (100 total)

Reading the diagram

Each box holds exactly one slip (a random permutation): Box $1 \to$ slip 3; Box $2 \to$ slip 6; Box $3 \to$ slip 1; Box $4 \to$ slip 5; ...

The 100 Prisoners Dilemma — Naive Strategy Fails

Strategy A: Open 50 Random Boxes

Each prisoner independently picks 50 boxes at random.

$P(\text{prisoner } i \text{ finds their number}) = \frac{50}{100} = \frac{1}{2}$

Since all 100 prisoners act **independently**:

$$P(\text{all 100 succeed}) = \left(\frac{1}{2}\right)^{100} \approx 8 \times 10^{-31} \approx \mathbf{0}$$

Is there a better strategy?

- The prisoners **cannot** change the odds for any individual prisoner beyond 1/2 (they must open exactly 50 out of 100 boxes).
- But they *can* **correlate** their choices using a shared strategy agreed upon beforehand.

The 100 Prisoners Dilemma — The Cycle Strategy

Strategy B: Follow Your Cycle

Prisoner i proceeds as follows:

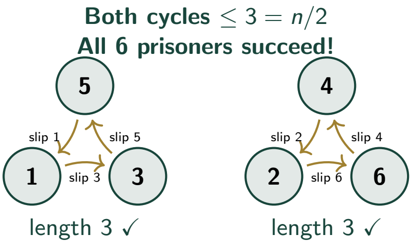
1. Open **Box** i (your own number).
2. Read slip j inside.
3. Open **Box** j.
4. Read the new slip, go to that box.
5. Repeat until you find **slip** i or you have opened 50 boxes.

Why this is clever

Every permutation decomposes into **disjoint cycles**. Following the chain starting at box i always leads back to slip i — if the cycle is short enough.

Worked example (6 prisoners):

Permutation: $1 \to 3 \to 5 \to 1$ and $2 \to 6 \to 4 \to 2$



The 100 Prisoners Dilemma — Probability Calculation

Key Counting Result

The number of permutations of $\{1, \ldots, 100\}$ that contain a cycle of length exactly k is:

$$\binom{100}{k}(k-1)! \cdot (100-k)! = \frac{100!}{k}$$

So $P(\text{there exists a cycle of length } k) = \dfrac{1}{k}$.

Computing the Success Probability

The strategy fails iff some cycle has length > 50 (at most one such cycle can exist, since $50+51 > 100$):

$$P(\text{failure}) = \sum_{k=51}^{100} \frac{1}{k} = \frac{1}{51} + \frac{1}{52} + \cdots + \frac{1}{100} \approx \ln 2 \approx 0.693$$

$$\boxed{P(\text{all 100 succeed}) = 1 - \ln 2 \approx \mathbf{30.7\%}}$$

Strategy	**Individual P**	**All succeed**
Random (50 boxes)	$1/2$	$\approx 10^{-30}$
Follow the cycle	$1/2$	$\approx$ **30.7%**

In-Class Activity – Lecture 2

Activity 2 – Set Theory & Counting

Part A – Sets: Let $S = \{1, 2, \dots, 10\}$, $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 3, 5, 7\}$.

(a) Find $A \triangle B$ (symmetric difference).

(b) Verify: $|A \cup B| = |A| + |B| - |A \cap B|$.

(c) Do A, $A^c \cap B$, $B^c \cap A^c$ partition S? Prove or disprove.

Part B – Counting:

(d) A lock has 3 dials each with digits 0–9. How many combinations?

(e) How many 3-letter words (A–Z) have no repeated letter?

(f) A president, VP, and secretary are chosen from 8 candidates. How many ways?

(g) **Bonus:** How many ways to arrange the letters in MISSISSIPPI?

Activity 2 – Solutions

Part A:

- (a) $A \cap B = \{3, 5, 7\}$; $A \triangle B = (A \cup B) - (A \cap B) = \{1, 2, 9\}$... actually $A \cup B = \{1, 2, 3, 5, 7, 9\}$, so $A \triangle B = \{1, 2, 9\}$
- (b) $|A| = 5, |B| = 4, |A \cap B| = 3$; $|A \cup B| = 6 = 5 + 4 - 3$✓
- (c) $A \cup (A^c \cap B) \cup (B^c \cap A^c) = A \cup B \cup (A \cup B)^c = S$✓; pairwise disjoint✓. **Yes.**

Part B:

- (d) $10^3 = \mathbf{1000}$
- (e) $P(26, 3) = 26 \times 25 \times 24 = \mathbf{15{,}600}$
- (f) $P(8, 3) = 8 \times 7 \times 6 = \mathbf{336}$
- (g) MISSISSIPPI: 11 letters; M:1, I:4, S:4, P:2. $\frac{11!}{1!4!4!2!} = \frac{39916800}{1152} = \mathbf{34{,}650}$